Introduction to Statistical Investigations

Second Edition

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Chapter 2.1

Generalization: How Broadly Do the Results Apply?

Sampling from a Finite Population: Proportions

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Warm-Up #1

- For the hot-dog-is-a-sandwich study, discuss with your neighbor:
  - What are the observational units?
  - What is the variable(s)?

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Statistic and Parameter

- So far in our study of tests of significance we've focused on *process* probabilities.
  - Is the probability that Buzz will push the correct button more than 0.5?
  - Is the probability that scissors will be thrown less than $\frac{1}{3}$?
  - Is it likely that less than 50% of the class will think a hot dog is a sandwich?

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Statistic and Parameter

- With Buzz, the **statistic** was the proportion of times Buzz pushed the correct button in the trials we looked at $\left(\frac{15}{16}\right)$ and the **parameter** was Buzz's probability (long-run proportion) of pushing the correct button.
- We used this sample **statistic** to tell us something about the **parameter**.

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Generalization

- Now we want to focus more on finite populations instead of an infinite process.
- Typically the entire population is not measured for what we are interested in and it is difficult to do so (this would be taking a **census**).

- We, therefore, take **samples** (some subgroup of the population) to give us information we want about a **population**.
- We want to **generalize** the information from the sample to the population.
- In this section, we will look at how to collect a sample that we feel is representative of the larger population.

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Example 2.1: Voter Turnout

- In 2018, the General Social Survey (GSS) asked if the respondent voted in the 2016 U.S. Presidential Election.
- Of the 2,181 eligible voters surveyed, 1,523 **(69.8%)** of them said yes.

- But the United States Election Project said there were 230,931,921 citizens eligible to vote in that election and only 136,753,936 **(59.2%)** of them voted.
- Why aren't these two percentages that represent voter turnout the same?

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Example 2.1: Population and Sample

- Perhaps the respondents to the GSS (the *sample*) were not representative of all eligible voters (the *population*).
- The **population** is the entire collection of observational units we are interested in, and the **sample** is a subset of the population on which we record data.
- What would convince you that GSS respondents are representative of the larger population?

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Example 2.1: Population and Sample

- Perhaps the respondents to the GSS (the *sample*) were not representative of all eligible voters (the *population*).
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- What would convince you that GSS respondents are representative of the larger population?
  - They need to come from all over the country.
  - Everyone in the voting-eligible population could potentially be selected for the survey.

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Example 2.1: Nonrandom Samples

- We wouldn't want to do a call-in survey, where only the proudest Americans replied.
- Or just sample from some convenient areas of the country.
- A **convenience sample** is a nonrandom sample of a population. **Voluntary response**, like a phone-in or mail-in survey, is a common type of convenience sample.
- Convenience samples often over sample certain segments of the population that can result in over or under-estimating the true population parameter.

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Example 2.1: Biased Sampling Method

- A sampling method is **biased** if statistics from samples consistently over or under-estimate the population parameter.
- Bias is a property of a sampling method, not the sample.
- A *method* must consistently produce non-representative results to be considered biased.

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Example 2.1: Simple Random Sample

- To prevent bias, we select a **random sample**, such as a *simple random sample* (SRS). How is this a better way of selecting a representative sample?
- A **simple random sample** ensures that every sample of size n is equally likely to be the sample selected.
- This also means that each observational unit in the population has an equal chance of being selected.

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Example 2.1: Sampling Frame

- A random sample is very different than “randomly asking your friends.”
- To create a simple random sample, you need a *sampling frame* (a list of all individuals in the population).
- Then have a computer randomly select names from the sampling frame.

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Example 2.1: General Social Survey

- Does the General Social Survey use a simple random sample?
- It is unlikely that the GSS has a complete sampling frame, such as a list of all adults in the United States.
- However, the GSS uses a method that tries to ensure that each U.S. adults has the same probability of being chosen as every other U.S. adult and thus makes the sample representative of the population.
- Why are the GSS results so different from the actual results? Was it just by random chance?

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Example 2.1: Sampling Students (1 of 13)

- To explore a whether the difference between a sample proportion and a population proportion is due to random chance, we need to explore distributions of sample proportions where we have access to the entire population.
- We will look at data collected from the registrar's office from a small college in the Midwest.
- Of the 2,919 students in this population, we know that **77.6%** of them lived on campus.

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Example 2.1: Sampling Students (2 of 13)

- We usually don't have information on an entire population like we do here.
- We usually need to make inferences about a population based on a sample.
- Suppose a researcher asks the first 30 students he finds on campus one morning whether they live on campus. This would be a quick and convenient way to get a sample.
- We suspect that this will lead to an overestimate of the proportion of students who live on campus. Why?

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Example 2.1: Sampling Students (3 of 13)

- To get a representative sample, we could take a simple random sample of 30 students from the college.
- Represent each student by ID numbers 1 to 2919
- Have the computer randomly select 30 numbers between 1 and 2,919
- The table on the next slide shows one such sample.

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Example 2.1: Sampling Students (4 of 13)

IDs of the 30 people selected, along with their residential status

ID	On campus?	ID	On campus?	ID	On campus?
827	Y	844	N	825	Y
1355	Y	90	Y	2339	N
1455	Y	1611	Y	2064	Y
2391	Y	2550	Y	2604	Y
575	Y	2632	Y	2147	Y
2049	N	2325	Y	2590	Y
895	N	2563	Y	1718	Y
1732	Y	1819	N	168	Y
2790	Y	968	Y	1777	Y
2237	Y	566	N	2077	Y

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Example 2.1: Sampling Students (5 of 13)

- What proportion live on campus?
  - $\hat{p}$ is the sample proportion
  - $\hat{p} = 0.80$
- The symbol that corresponds to $\hat{p}$ but describes the population proportion, not the sample proportion, is π

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Example 2.1: Sampling Students (6 of 13)

- How do we know if $\hat{p}$ is close to the population value?
- A different sample of 30 students would probably have had a different value
- How are these statistics useful in estimating the population parameter values?
- Let's take more simple random samples of 30 students to examine the null distribution of the statistics from other samples.

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Example 2.1: Sampling Students (7 of 13)

- We took 5 different SRS s of 30 students
- Each sample gives different statistics
- This is called **sampling variability**
- The statistics from each sample don't change much:
  - Sample proportions from 0.63 to 0.83

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Example 2.1: Sampling Students (8 of 13)

Random sample	**1**	**2**	**3**	**4**	**5**
proportion on campus($\hat{p}$)	0.80	0.83	0.77	0.63	0.83

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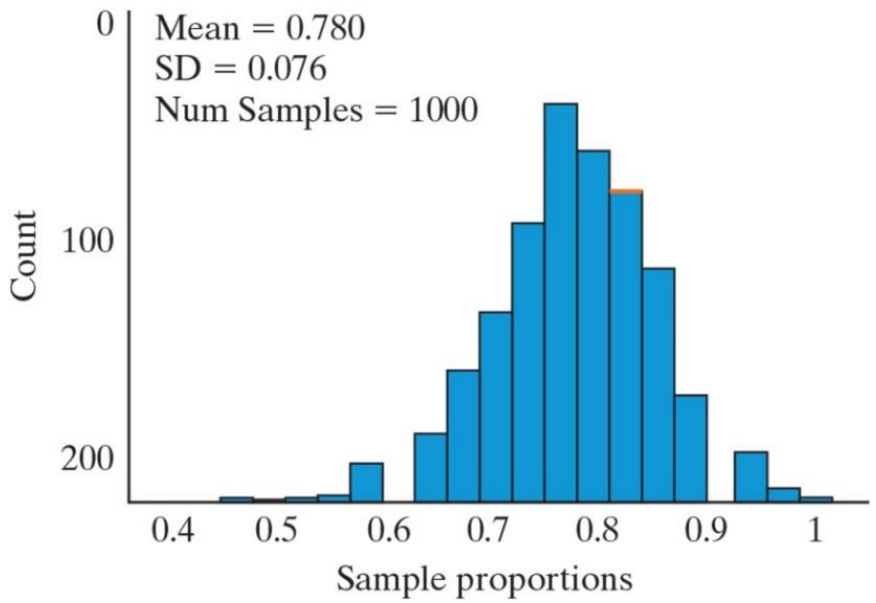
Example 2.1: Sampling Students (9 of 13)

- We took 1,000 simple random samples and the graph below is of the 1,000 sample proportions (for living on campus).
- The mean of the distribution falls near the population parameter



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Example 2.1: Sampling Students (10 of 13)

- What would happen if we took all possible random samples of 30 students from this population?
  - The average of the sample proportions would match the population proportion exactly
- Statistics computed from simple random samples cluster around the parameter
  - There is no tendency to over or underestimate the parameter
- Simple random sampling is an unbiased method of sampling for a sample proportion

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Example 2.1: Sampling Students (11 of 13)

- We can generalize when we use simple random sampling because it creates:
  - A sample that is representative of the population
  - A sample statistic that is close to the parameter

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Example 2.1: Sampling Students (12 of 13)

- But what about the standard deviation?
- When the population size is large, we can predict the sample-to-sample variation in sample proportions:

$$SD(\hat{p}) = \sqrt{\frac{\pi(1-\pi)}{n}}$$

- For this example, we get:

$$SD(\hat{p}) = \sqrt{\frac{0.776(1-0.776)}{30}} = 0.076$$

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Example 2.1: Sampling Students (13 of 13)

- When sampling from a large population, the standard deviation of sample proportions is estimated by the formula, $SD(\hat{p}) = \sqrt{\frac{\pi(1-\pi)}{n}}$ where π is the proportion of success in the population and n is the sample size.
- The population is considered large enough when it is more than 20 times the sample size.

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Example 2.1: The Central Limit Theorem

- The **Central Limit Theorem for sampling from a large finite population** says that the distribution of sample proportions from repeated random samples will be approximately normal if there are at least 10 successes and at least 10 failures in each sample.

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Example 2.1: Back to Voter Turnout

- The observed sample proportion was $\hat{p} = 0.698$, with a sample size of $n = 2{,}181$.
- We would like to make inferences about the parameter of interest, π, the proportion of all eligible voters in 2016 who voted in the presidential election.
- Is it plausible that we would observe a sample proportion of 0.698 if the population proportion is equal to 0.592?

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Example 2.1: Using the Central Limit Theorem

- We can assume the distribution of sample proportions is approximately normal. (Why?)
  - With of mean of 0.592
  - And SD of $\sqrt{\dfrac{0.592(1-0.592)}{2{,}181}} = 0.0105$

- This means the sample proportion from the GSS has a standardized statistic of:

$$z = \frac{0.698 - 0.592}{0.0105} = 10.10$$

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Example 2.1: Strong Evidence

- With a standardized statistic of 10.10, we have very strong evidence that this difference did not arise from random chance alone.
- What could be another explanation for this difference between sample proportion of voters and population proportion of voters?

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Example 2.1: Nonsampling Concerns (1 of 2)

- When a question is posed to someone in a poll that has a *socially desirable* response (like you voted), individuals will sometimes overestimate or incorrectly recall their past behavior.
- In this case, we have reason to believe that the difference between 0.698 and 0.592 is not due to a badly collected sample (a "sampling concern") or due to random chance, but rather due to individuals not being honest in their responses.
- This is a type of **nonsampling concern**.

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Example 2.1: Nonsampling Concerns (2 of 2)

- In addition to carefully selecting the sample, you need to guard against other possible sources of bias as well.
- For example, when talking to people, their memories and truthfulness can impact their answers.
- Other examples of nonsampling concerns include:
  - The effects that the wording of a question can have on responses
  - The effects that interviewers can have by virtue of their demeanor, sex, race and other characteristics
  - Using an uncalibrated scale, etc.

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Cool-Down #1

- 1.5.1: Which long-run proportion of success, π, gives the largest standard deviation of the null distribution when the sample size is 10?
  - A. 0.05 B. 0.25 C. 0.50 D. 0.90

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Cool-Down #2

- 1.4.5: According to the National Coffee Drinking Study from the National Coffee Association, 40% of 18-24 year-olds in the USA regularly drink coffee every day. Suppose this number is accurate. In a sample of 140 students at Hope College, we found that 63 of them (45%) drink coffee regularly. We want to test whetehr the daily coffee-drinking habits of students at Hope College differ from the 40% claim for the nation. Which of A-E is the appropriate sample statistic and proper symbol?
- A. $\hat{p} = 0.40$ B. $\pi = 0.40$ C. $\pi = 0.45$ D. $n = 140$

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Cool-Down #3

- 2.1.8: The administration at a local high school is considering building a new gym. In order to find out how the student body feels about this, a student organization decides to conduct a survey of the students. Which of the sampling methods in A-D is a simple random sample from all the students at the school?
  - A. Survey random students who are leaving after school sports practices.
  - B. Randomly select 40 students in each grade level and survey the selected students.
  - C. Randomly select two homeroom classes from the 50 homeroom classes at the school and survey all students from those two classes.
  - D. Obtain a list of all students attending the high school, randomly select students from the list, and survey the students selected.

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